

THE TWO-COLUMN RULE (TC) FOR $e_k(Y) \bullet (e_a e_b)$, PROVED FOR ALL k : OUTER PEEL, A THREE-POLE RESIDUE STEP, AND A TWO-INDEX CLOSING IDENTITY

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ABSTRACT. For every number of variables $m \geq 0$ and every $k \geq 0$ we prove a closed generating-function rule (TC) for $\sum_{a,b} z^a w^b t^{-\binom{k}{2}} e_k(Y) \bullet (e_a e_b)$, as an identity in $\Lambda_m \otimes \mathbb{Q}(s, t)(z, w)$, $s = q^{-1}$. The right side is a sum of chain shifts $e_b E(t^i z) E(t^j w)$ weighted by products of two one-column weights $N_i^{(n_1)} N_j^{(n_2)}$ and a cross kernel K_{ij} . Via Hikita's Definition 3.4 and Lemma 3.3 (arXiv:2503.23597), the coefficients are $e_k \star (e_a e_b)$. The proof is the Day 207b outer-peel argument; the closing identity reduces to a three-line polynomial identity.

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 notes/2026-09-26-ek-star-er-proved-for-clio.pdf, commit e98fb0b.

Please check in particular: §3, the step map, especially the Laurent/residue extraction for the e_n coefficient; and §5, Step B, the normalisations. Items **(P1)**–**(P8)** are machine-checked, and the step map was compared against an independent symbolic recursion.

1. CONVENTIONS AND STATEMENT

The conventions are those of Day 207b (“207b” below; source file proofs/2026-09-26-day207b-ek-star-er-general-k-P-
 • $s = q^{-1}$, $a_{ij} = (X_i - tX_j)/(X_i - X_j)$, $E(z) = \prod_i (1 + X_i z) = \sum_r e_r z^r$;
 • $\alpha_j = \prod_{i \leq j} (s - t^i)/(t; t)_j$;
 • $c(n, j) = \frac{(s; t)_{n-j}}{(t; t)_{n-j}} (\alpha_j - s t^{n-j} \alpha_{j-1})$;
 • $N_j^{(n)} = t^{-nj} c(n, j)$.
 Define the cross kernel

$$K_{ij} := \prod_{p < i} \frac{t^p z - s w}{t^p z - t^j w} \prod_{r < j} \frac{s z - t^r w}{t^i z - t^r w},$$

and, for $i, j \geq 0$,

$$V_{ij}^{(n)} := K_{ij} \sum_{n_1 + n_2 = n} s^{(n_1 - i) + (n_2 - j)} t^{-(n_1 - i)j - (n_2 - j)i} N_i^{(n_1)} N_j^{(n_2)} z^{-n_1} w^{-n_2},$$

with $V := 0$ if $i < 0$ or $j < 0$. Since $c(n, j) = 0$ for $n < j$, the constraints $n_1 \geq i$ and $n_2 \geq j$ are automatic.

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Theorem 1 (TC). *For all $m \geq 0$ and $k \geq 0$, in $\Lambda_m \otimes \mathbb{Q}(s, t)(z, w)$,*

$$\Gamma_k := \sum_{a,b} z^a w^b t^{-\binom{k}{2}} e_k(Y) \bullet (e_a e_b) = T_k := \sum_{b,i,j} s^{2b} t^{-b(i+j)} V_{ij}^{(k-b)} e_b E(t^i z) E(t^j w).$$

Writing $b_0 = b$ and $n_1 + n_2 = k - b$, this is PROVE.md's (TC) verbatim.

Corollary 2. Γ_k is a polynomial in z and w . Hence all the poles of T_k at $z = t^c w$ and at $z = 0$ cancel in the total, and the coefficient of $z^a w^b$ in T_k is $e_k \star (e_a e_b)$, in Hikita's reading (207b §0). In particular this gives $e_2 \star (e_a e_b)$ for all a, b .

2. THE RECURSION (R2)

This is 207b §§2–3 verbatim. (A_k) and (K_k) hold for any symmetric F , and $F = e_a e_b$ is symmetric. With

$$\varphi(u) := \frac{u(1+su z)(1+su w)}{(1+uz)(1+uw)} \quad \text{we get} \quad \Gamma_k = E(z)E(w) \sum_{|A|=k} \prod_A^\times \prod_{a \in A} \varphi(X_a).$$

The reason is that the generating function of $(\pi^k(e_a e_b))^{(A)} = X_A e_a(X_{A^c}, sX_A) e_b(X_{A^c}, sX_A)$ is $X_A \prod_{A^c} (1 + Xz)(1 + Xw) \prod_A (1 + sXz)(1 + sXw)$.

The proof of 207b (R) (ordered formula (C2), then Poincaré (C3), then separating $a_1 = i$) uses nothing about φ . Together with $\varphi(X_i)E(z)E(w) = X_i(1 + sX_i z)(1 + sX_i w)E(\hat{X}_i; z)E(\hat{X}_i; w)$, it gives

$$(R2) \quad [k] \Gamma_k(X) = \sum_i X_i (1 + sX_i z)(1 + sX_i w) \prod_{j \neq i} a_{ij} \Gamma_{k-1}(\hat{X}_i), \quad m \geq 1.$$

For $m = 0$, $\Gamma_k = 0$ when $k \geq 1$.

3. STEP MAP (RATIONAL FORM OF LEMMA 2)

Lemma 3 (Lemma 2', residues). *Let $K \supseteq \mathbb{Q}(t)$ be a field and $H \in K(y)$ with $H(0) = 0$. Suppose the poles of H/y are simple, lie at points $p_\ell \notin \{X_i\}$, and that $H(y)/y = O(1/y)$ at ∞ . Then in $K(X_1, \dots, X_m)$,*

$$(1-t) \sum_i H(X_i) \prod_{j \neq i} a_{ij} = - \sum_\ell \text{Res}_{y=p_\ell} \left[\frac{H(y)Q(1/y)}{y} \right] - \text{Res}_{y=\infty} \left[\frac{H(y)Q(1/y)}{y} \right],$$

where $Q(1/y) := \prod_j (y - tX_j)/(y - X_j)$.

Proof. Let $f(y) := H(y)Q(1/y)/y$. It is regular at $y = 0$. Its residue at $y = X_i$ is $(1-t)H(X_i) \prod_{j \neq i} a_{ij}$. The sum of all residues of a rational function is 0. $\square$

This is the same residue argument as Day 205b Lemma 2 (Macdonald III (2.10)); Lemma 2 is the case $H = y^n$. We use it with $K = \mathbb{Q}(s, t, z, w, x)$. At a pole $y = -1/a$ we have $Q(-a) = E(ta)/E(a)$ exactly, so no power series are needed.

Step map. Fix one term $\mathcal{W} \cdot e_b(\hat{X}_i)E(\hat{X}_i; \gamma)E(\hat{X}_i; \delta)$ of $T_{k-1}(\hat{X}_i)$, where $\gamma = t^i z$, $\delta = t^j w$, and $\mathcal{W} \in \mathbb{Q}(s, t)(z, w)$. Put $u = X_i$. Then

- $e_b(\hat{X}_i) = \sum_{c \leq b} (-u)^c e_{b-c} = [x^b] E(x)/(1+xu)$;
- $E(\hat{X}_i; a) = E(a)/(1+au)$.

So, after multiplying by $u(1+su z)(1+su w)$ and summing over i against $\prod a_{ij}$, the term contributes

$$\mathcal{W} E(\gamma)E(\delta) \cdot [x^b] \sum_i E(x) H(X_i; x) \prod_{j \neq i} a_{ij}, \quad H(y; x) := \frac{y(1+sy z)(1+sy w)}{(1+\gamma y)(1+\delta y)(1+xy)}.$$

Here $[x^b]$ is taken of a finite sum of rational functions regular at $x = 0$, so it commutes with the sum over i . The function H/y has simple poles at $-1/a$ for $a \in \{x, \gamma, \delta\}$, and it is $\sim c/(xy)$ at ∞ , where

$$c := \frac{s^2 zw}{\gamma \delta} = s^2 t^{-i-j}.$$

The residues are $\rho_a := \frac{(a-sz)(a-sw)}{a \prod_{a' \neq a} (a-a')}$. Lemma 3 then gives

$$(1-t) \sum_i H(X_i; x) \prod a_{ij} = \frac{c}{x} - \sum_{a \in \{x, \gamma, \delta\}} \rho_a \frac{E(ta)}{E(a)}.$$

Multiply by $E(x)E(\gamma)E(\delta)$ and write $E(x) = \sum e_n x^n$ and $E(tx) = \sum t^n e_n x^n$. The coefficient of e_n becomes $[x^{b-n}]$ of the Laurent expansion at $x = 0$ of

$$\tilde{\rho}_n(x) = \left(\frac{c}{x} - t^n \rho_x \right) E(\gamma)E(\delta) - \rho_\gamma E(t\gamma)E(\delta) - \rho_\delta E(\gamma)E(t\delta).$$

This function has simple poles at $x \in \{0, \gamma, \delta\}$ and is $O(1/x)$ at ∞ . So:

- $n = b + 1$: the coefficient is $\text{Res}_0 \tilde{\rho}_{b+1} = c(1 - t^{b+1})E(\gamma)E(\delta)$, since $\text{Res}_0 \rho_x = c$.
- $n \leq b$: there is no residue at ∞ . With $\text{Res}_\gamma \rho_x = \kappa_\gamma$ and $\text{Res}_\gamma \rho_\gamma = -\kappa_\gamma$, the coefficient is

$$-\kappa_\gamma \gamma^{n-b-1} (E(t\gamma) - t^n E(\gamma))E(\delta) - \kappa_\delta \delta^{n-b-1} E(\gamma)(E(t\delta) - t^n E(\delta)),$$

$$\text{where } \kappa_\gamma := \frac{(\gamma-sz)(\gamma-sw)}{\gamma(\gamma-\delta)} \text{ and } \kappa_\delta := \frac{(\delta-sz)(\delta-sw)}{\delta(\delta-\gamma)}.$$

- $n > b + 1$: the coefficient is 0.

For one column (w -free) this collapses to 207b (L)(a)+(b). `step_map_check.py` confirms that this closed step map equals Day 208's symbolic `gf_recursion.step`.

4. STEP LEMMA (L2) AND THE REDUCTION TO (★2)

Lemma 4 (L2). *For all $m \geq 0$ and $k \geq 1$:*

$$\sum_i X_i (1 + sX_i z)(1 + sX_i w) \prod_{j \neq i} a_{ij} T_{k-1}(\hat{X}_i) = [k] T_k(X).$$

Proof. Write $T_{k-1} = \sum W'_{bij} e_b E(t^i z) E(t^j w)$ with $W'_{bij} = s^{2b} t^{-b(i+j)} V_{ij}^{(k-1-b)}$. By §3, $(1-t) \cdot \text{LHS}$ is a sum of terms $e_{b'} E(t^i z) E(t^j w)$. For each (b', i, j) the coefficient is

$$\begin{aligned} & W'_{b'-1, ij} c (1 - t^{b'}) + \sum_{b \geq b'} \left[t^{b'} W'_{bij} (\kappa_\gamma \gamma^{b'-b-1} + \kappa_\delta \delta^{b'-b-1}) \right. \\ & \quad \left. - W'_{b, i-1, j} \kappa_\gamma^{(i-1, j)} (t^{i-1} z)^{b'-b-1} - W'_{b, i, j-1} \kappa_\delta^{(i, j-1)} (t^{j-1} w)^{b'-b-1} \right]. \end{aligned}$$

Here $\kappa^{(i-1, j)}$ denotes κ evaluated at $(\gamma, \delta) = (t^{i-1} z, t^j w)$. The two last terms are the shifted outputs $E(t\gamma')$ of the $(i-1, j)$ and $(i, j-1)$ inputs, and they land on $E(t^i z)E(t^j w)$.

Put $n = k - b'$ and $p = b - b'$, and factor out $s^{2b'} t^{-b'(i+j)}$. The prefix weights give:

- $W'_{b'-1, ij} \cdot c = s^{2b'} t^{-b'(i+j)} V_{ij}^{(n)}$;
- $W'_{bij} = s^{2b'} t^{-b'(i+j)} c^p V_{ij}^{(n-1-p)}$;
- $W'_{b, i-1, j} = s^{2b'} t^{-b'(i+j)} t^{b'} (ct)^p V_{i-1, j}^{(n-1-p)}$, and the same for $(i, j-1)$.

So the coefficient is $s^{2b'}t^{-b'(i+j)}[(1-t^{b'})V_{ij}^{(n)} + t^{b'} \cdot \Sigma]$, where

$$\Sigma := \sum_{p \geq 0} c^p \left[V_{ij}^{(n-1-p)} (\kappa_\gamma \gamma^{-p-1} + \kappa_\delta \delta^{-p-1}) - t^p V_{i-1,j}^{(n-1-p)} \kappa_\gamma^{(i-1,j)} (t^{i-1} z)^{-p-1} \right. \\ \left. - t^p V_{i,j-1}^{(n-1-p)} \kappa_\delta^{(i,j-1)} (t^{j-1} w)^{-p-1} \right].$$

(★2) For all $n \geq 0$ and $i, j \geq 0$: $\Sigma = (1-t^n)V_{ij}^{(n)}$. (Proved in §5.)

Given (★2), the coefficient is $s^{2b'}t^{-b'(i+j)}(1-t^{b'+n})V_{ij}^{(n)} = (1-t^k)W_{b'ij}^{(k)}$. Divide by $(1-t)$. This matches T_k term by term, so the sums agree and no uniqueness of the representation is needed. $\square$

Proof of Theorem 1 from (L2). Induct on k , for all m at once. For $k = 0$, $\Gamma_0 = E(z)E(w) = T_0$, since $V_{00}^{(0)} = 1$. Let $k \geq 1$.

- $m = 0$: $\Gamma_k = 0$, and (L2) with its empty left side gives $[k]T_k = 0$.
- $m \geq 1$: by (R2), the induction hypothesis in $\hat{X}_i$, and (L2), $[k]\Gamma_k = [k]T_k$.
(This is 207b's (E_k) argument word for word.) $\square$

5. PROOF OF (★2)

Generating function. Let $\mathcal{V}_{ij}(x) := \sum_n V_{ij}^{(n)} x^n$. With $C_i(x) := \sum_n c(n, i) x^n$ (207b), the sum over (n_1, n_2) factors as

$$\mathcal{V}_{ij}(x) = K_{ij} s^{-i-j} t^{2ij} C_i(X) C_j(W), \quad X := st^{-i-j} x/z, \quad W := st^{-i-j} x/w.$$

The t -exponent is $-(n_1 - i)j - n_1 i - (n_2 - j)i - n_2 j = -(n_1 + n_2)(i + j) + 2ij$.

Multiply (★2) by x^n and sum over n (the p -sums are convolutions). Then (★2) for all n is equivalent to

$$(\star 2\text{-GF}) \quad \mathcal{V}_{ij}(x)U(x) - \mathcal{V}_{ij}(tx) = -\kappa_\gamma^{(i-1,j)} \frac{x}{t^{i-1}z - ctx} \mathcal{V}_{i-1,j}(x) - \kappa_\delta^{(i,j-1)} \frac{x}{t^{j-1}w - ctx} \mathcal{V}_{i,j-1}(x),$$

where $U := 1 - \frac{\kappa_\gamma x}{\gamma - cx} - \frac{\kappa_\delta x}{\delta - cx}$.

Step A: U is a product. Let $F(y) := \frac{(y-sz)(y-sw)}{y(y-\gamma)(y-\delta)}$.

- Its partial fractions are $F = \frac{c}{y} + \frac{\kappa_\gamma}{y-\gamma} + \frac{\kappa_\delta}{y-\delta}$. (P1)

- Hence $xF(cx) = 1 - \frac{\kappa_\gamma x}{\gamma - cx} - \frac{\kappa_\delta x}{\delta - cx} = U$. (P2)

- Now evaluate F at $y = cx$. Using $c\gamma\delta = s^2zw$, together with $swx/(\gamma\delta) = X$, $szx/(\gamma\delta) = W$ and $s^2zwx/(\gamma^2\delta) = sX/A$ ($A := t^i$, $B := t^j$), we get

$$(P3) \quad U = \frac{(1-X)(1-W)}{(1-sX/A)(1-sW/B)}.$$

This is the two-column version of 207b's move $C_j(x)[1 - (1-st^{-j})x/(1-st^{-j}x)] = C_j(x)(1-x)/(1-st^{-j}x)$. The residue function F is that move.

Step B: normalise. From 207b §4, $C_i(x) = x^i G(tx) [\alpha_i(1-sx)/(1-x) - s\alpha_{i-1}]$. Together with the linear factorization $\alpha_i(1-sy) - s\alpha_{i-1}(1-y) = D_i(1-st^{-i}y)$, this gives

$$C_i(x) = D_i x^i G(tx) \frac{1-sx/A}{1-x}, \quad \frac{G(y)}{G(ty)} = \frac{1-sy}{1-y}, \quad (1-t^i)D_i = t(s-t^{i-1})D_{i-1} \quad (i \geq 1).$$

Divide (★2-GF) by $K_{ij}s^{-i-j}t^{2ij} \cdot D_i D_j X^i W^j G(t^2 X) G(t^2 W)$. This is legitimate because $D_i \neq 0$ in $\mathbb{Q}(s, t)$. The normalised pieces are:

- $\hat{C}(X) := C_i(X)/(D_i X^i G(t^2 X)) = \frac{(1-stX)(1-sX/A)}{(1-tX)(1-X)}$.
- $\hat{C}_t(X) := C_i(tX)/(\dots) = \frac{A-stX}{1-tX}$.
- **Left side.** $L_0 := \hat{C}(X)\hat{C}(W)U - \hat{C}_t(X)\hat{C}_t(W)$. By (P3),

$$L_0 = \frac{(1-stX)(1-stW) - (A-stX)(B-stW)}{(1-tX)(1-tW)}.$$

- **First shifted term** ($i \geq 1$). Use:

$$- \frac{x}{t^{i-1}z - ctx} = \frac{tBX/s}{1-st^2X/A}; \quad (\text{P4})$$

- $\mathcal{V}_{i-1,j}$ has prefactor $s \cdot t^{-2j} \cdot K_{i-1,j}/K_{ij}$ relative to $\mathcal{V}_{ij}$;
- $C_{i-1}(tX)/(D_i X^i G(t^2 X)) = \frac{1-A}{ts-A} \cdot \frac{A}{t} X^{-1} \frac{1-st^2X/A}{1-tX}$, in which the factor $(1-st^2X/A)$ cancels the denominator of (P4);
- the telescoped ratio, with $\rho := z/w = W/X$,

$$(P8) \quad \frac{K_{i-1,j}}{K_{ij}} = \frac{B(A\rho/t - B)(A\rho/t - 1/t)}{(A\rho/t - s)(A\rho/t - B/t)};$$

- the product with $\kappa_\gamma^{(i-1,j)} = \frac{(A/t - s)(A\rho/t - s)}{(A/t)(A\rho/t - B)}$, which is

$$(P6) \quad \frac{B(A-st)(A\rho-1)}{A(A\rho-B)}.$$

The first shifted term contributes $-S_1$ to the right side of ($\star$ 2-GF), with

$$S_1 = - \frac{(1-A)(B-stW)(AW-X)}{(AW-BX)(1-tX)(1-tW)}.$$

For $i = 0$ the term is absent, and the formula has the factor $(1-A) = 0$ anyway.

- **Second shifted term.** By the $z \leftrightarrow w$, $i \leftrightarrow j$ symmetry of K_{ij} , V and κ , it contributes $-S_2$ with

$$S_2 = - \frac{(1-B)(A-stX)(W-BX)}{(AW-BX)(1-tX)(1-tW)}.$$

Step C: the closing polynomial identity. ($\star$ 2-GF) is now $L_0 + S_1 + S_2 = 0$. Put $u := stX$ and $v := stW$. We must show

$$(P7) \quad (AW-BX)[(1-u)(1-v) - (A-u)(B-v)] = (1-A)(B-v)(AW-X) + (1-B)(A-u)(W-BX).$$

- The bracket on the left is $(1-AB) - (1-B)u - (1-A)v$.
- *Constant parts.* $(1-A)B(AW-X) + (1-B)A(W-BX) = AW - BX - A^2BW + AB^2X = (1-AB)(AW-BX)$. This matches.
- *v-parts.* The right has $-(1-A)v(AW-X)$ and the left has $-(1-A)v(AW-BX)$. They differ by $(1-A)(1-B)vX$.
- *u-parts.* The right has $-(1-B)u(W-BX)$ and the left has $-(1-B)u(AW-BX)$. They differ by $-(1-A)(1-B)uW$.
- The total difference is $(1-A)(1-B)(vX - uW) = (1-A)(1-B) \cdot st(WX - XW) = 0$.

($\star$ 2-GF) is therefore an identity of rational functions in x with $A = t^i$ and $B = t^j$. None of the denominators used vanishes identically: $(AW-BX) \propto (t^i z - t^j w)$, and $(1-sX/A)$, $(ts-A)$, $(A\rho/t - s)$ are all nonzero. Comparing coefficients of x^n gives ($\star$ 2) for every $n \geq 0$ (at $n = 0$ it reads $0 = 0$), and this completes the proof of Theorem 1. $\square$

6. REMARKS

What made it short.

- *Same skeleton as 207b*. Outer peel, then one Lemma-2 evaluation per step, then an index identity. Only the evaluation has three poles instead of two, and the index identity has two indices.
- *Rational Lemma 2'*. Doing Lemma 2 as a residue identity on $K(y)$ avoids all power-series book-keeping. The (w, γ) difference quotient of 207b becomes a residue sum over $\{\gamma, \delta\}$.
- *The cross poles*. K_{ij} is forced by the κ 's and needs no separate derivation. The whole two-index recursion collapses once U factors as $(1 - X)(1 - W)/((1 - sX/A)(1 - sW/B))$, via the residue function F , and the kernel ratio $\times \kappa$ collapses to $(AW - X)/(AW - BX)$. What remains is (P7), which says in effect that “ $u/X = v/W$ ”.
- *No q -Pfaff-Saalschütz*. PROVE.md expected one; it is not needed. The two columns interact only through the rational factor $(AW - X)/(AW - BX)$, and each column keeps its own 207b generating function C_i .

Verification. (scripts/day209/, all logs ALL OK).

check						scope	log
step	map	(§3)	=	Day	208	symbolic, $k \leq 3$	step_map_check.log
gf_recursion.step							
iterated step map = (TC)						symbolic, $k \leq 4$	step_map_check.log
(★2-GF) with free A, B						symbolic, exact 0	star2_rational.py
(★2) coefficient form						$n \leq 8$, all $i + j \leq n$, 3 exact random points	star2_coeff_check.log
(P1)–(P8)						symbolic	check_writeup_steps.log
(TC) vs direct AHA (Day 208)						$m \leq 6$, $k \leq 4$, and $(6, 5), (7, 4), (7, 5)$	scripts/day208/ check_conj*.log

Scope. Operator identity in $\Lambda_m \otimes \mathbb{Q}(s, t)(z, w)$, for every $m \geq 0$ and $k \geq 0$.

Gaps. None in the argument. The load-bearing inputs are 207b's (A_k) , (K_k) and (R), which were proved and re-used verbatim; 207b's C_i closed form and D -recursion; and the residue theorem. The $\star$ -reading uses Hikita 2503.23597 Def. 3.4 / Lemma 3.3 (verified-quote), as in 207b.

Not claimed.

- An explicit bounded coefficient formula for $e_2 \star (e_a e_b)$. By Day 208 §2 there is none of bounded length: the tail $T(q) + T(N - q) = 0$ comes from the cross poles.
- DS at length 3 for general k . Extracting it from (TC) is a separate job and is not claimed here. Novelty. As per novelty audit 12 (Day 208); no browsing was done on Day 209. The kernel (K_k) is classical (Macdonald III (2.2)).

Next. The ℓ -column rule for $\sum z_1^{a_1} \cdots z_\ell^{a_\ell} e_k(Y) \bullet (e_{a_1} \cdots e_{a_\ell})$. The same proof should go through with $F(y) = \prod_c (y - sz_c)/(y \prod_c (y - \gamma_c))$, and it would give $e_k \star e_\lambda$ for all λ . Conjecture (not proved here): $U = \prod_c (1 - X_c)/(1 - sX_c/A_c)$, with pairwise cross kernels.